

# Artificial Intelligence

| Course Outcomes |                                                                                                                                          |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>C01</b>      | Understanding about concept and definition AI along with its agents , its structure and environments (L2)                                |
| <b>C02</b>      | Apply various AI search algorithms (uninformed, informed, heuristic, constraint satisfaction, genetic algorithms). (L3)                  |
| <b>C03</b>      | Apply the reinforcement learning and natural language processing in AI. (L3)                                                             |
| <b>C04</b>      | Apply the natural language for communication and understand the perception of image formation, reconstruction of 3D world etc., (L3, L2) |
| <b>C05</b>      | Understand the Robotic perception like planning to move etc., and to understand the Philosophical foundation of AI. (L2)                 |

# ARTIFICIAL INTELLIGENCE

## UNIT-I

### What is artificial intelligence?

1. Artificial Intelligence is the branch of computer science concerned with making computers behave like humans.
2. Major AI textbooks define artificial intelligence as "the study and design of Intelligent agents," where an intelligent agent is a system that perceives its environment and takes actions which maximize its chances of success.
3. John McCarthy, who coined the term in 1956, defines it as "the science and engineering of making intelligent machines, especially intelligent computer programs."
4. The definitions of AI according to some text books are categorized into approaches and are summarized in the table below :

|                                                                                                                                                                    |                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Systems that think like humans</b><br>"The exciting new effort to make computers think ... machines with minds, in the full and literal sense."(Haugeland,1985) | <b>Systems that think rationally</b><br>"The study of mental faculties through the use of computer models."<br>(Charniak and McDermont,1985) |
| <b>Systems that act like humans</b><br>The art of creating machines that performs functions that require intelligence when performed by people."(Kurzweil,1990)    | <b>Systems that act rationally</b><br>"Computational intelligence is the study of the design of intelligent agents."(Poole et al.,1998)      |
|                                                                                                                                                                    |                                                                                                                                              |

### Goals of AI

- To Create Expert Systems – The systems which exhibit intelligent behavior, learn, demonstrate, explain, and advice its users.
- To Implement Human Intelligence in Machines – Creating systems that understand, think, learn, and behave like humans.

### Applications of Artificial Intelligence:

1. **Gaming** – AI plays crucial role in strategic games such as chess, poker, tic-tac-toe, etc., where machine can think of large number of possible positions based on heuristic knowledge.

2. **Natural Language Processing** – It is possible to interact with the computer that understands natural language spoken by humans.
3. **Expert Systems** – There are some applications which integrate machine, software, and special information to impart reasoning and advising. They provide explanation and advice to the users.
4. **Vision Systems** – These systems understand, interpret, and comprehend visual input on the computer. For example,
  5. A spying aeroplane takes photographs, which are used to figure out spatial information or map of the areas.
  6. Doctors use clinical expert system to diagnose the patient.
  7. Police use computer software that can recognize the face of criminal with the stored portrait made by forensic artist.
8. **Speech Recognition** – Some intelligent systems are capable of hearing and comprehending the language in terms of sentences and their meanings while a human talks to it. It can handle different accents, slang words, noise in the background, change in human's noise due to cold, etc.
9. **Handwriting Recognition** – The handwriting recognition software reads the text written on paper by a pen or on screen by a stylus. It can recognize the shapes of the letters and convert it into editable text.
10. **Intelligent Robots** – Robots are able to perform the tasks given by a human. They have sensors to detect physical data from the real world such as light, heat, temperature, movement, sound, bump, and pressure. They have efficient processors, multiple sensors and huge memory, to exhibit intelligence. In addition, they are capable of learning from their mistakes and they can adapt to the new environment.

## History of AI

### Maturation of Artificial Intelligence (1943-1952)

1. **Year 1943:** The first work which is now recognized as AI was done by Warren McCulloch and Walter Pitts in 1943. They proposed a model of **artificial neurons**.
2. **Year 1949:** Donald Hebb demonstrated an updating rule for modifying the connection strength between neurons. His rule is now called **Hebbian learning**.
3. **Year 1950:** The Alan Turing who was an English mathematician and pioneered Machine learning in 1950. Alan Turing publishes "**Computing Machinery and Intelligence**" in which he proposed a test. The test can check the machine's ability to exhibit intelligent behavior equivalent to human intelligence, called a **Turing test**.

### The birth of Artificial Intelligence (1952-1956)

1. **Year 1955:** An Allen Newell and Herbert A. Simon created the "first artificial intelligence program" which was named as "**Logic Theorist**". This program had proved 38 of 52 Mathematics theorems, and find new and more elegant proofs for some theorems.
2. **Year 1956:** The word "Artificial Intelligence" first adopted by American Computer scientist John McCarthy at the Dartmouth Conference. For the first time, AI coined as an academic field.

At that time high-level computer languages such as FORTRAN, LISP, or COBOL were invented. And the enthusiasm for AI was very high at that time.

### The golden years-Early enthusiasm (1956-1974)

1. **Year 1966:** The researchers emphasized developing algorithms which can solve mathematical problems. Joseph Weizenbaum created the first chatbot in 1966, which was named as ELIZA.
2. **Year 1972:** The first intelligent humanoid robot was built in Japan which was named as WABOT-1.

### The first AI winter (1974-1980)

1. The duration between years 1974 to 1980 was the first AI winter duration. AI winter refers to the time period where computer scientist dealt with a severe shortage of funding from government for AI researches.
2. During AI winters, an interest of publicity on artificial intelligence was decreased.

### A boom of AI (1980-1987)

1. **Year 1980:** After AI winter duration, AI came back with "Expert System". Expert systems were programmed that emulate the decision-making ability of a human expert.
2. In the Year 1980, the first national conference of the American Association of Artificial Intelligence **was held at Stanford University**.

### The second AI winter (1987-1993)

1. The duration between the years 1987 to 1993 was the second AI Winter duration.
2. Again Investors and government stopped in funding for AI research as due to high cost but not efficient result. The expert system such as XCON was very cost effective.

### The emergence of intelligent agents (1993-2011)

1. **Year 1997:** In the year 1997, IBM Deep Blue beats world chess champion, Gary Kasparov, and became the first computer to beat a world chess champion.
2. **Year 2002:** for the first time, AI entered the home in the form of Roomba, a vacuum cleaner.
3. **Year 2006:** AI came in the Business world till the year 2006. Companies like Facebook, Twitter, and Netflix also started using AI.

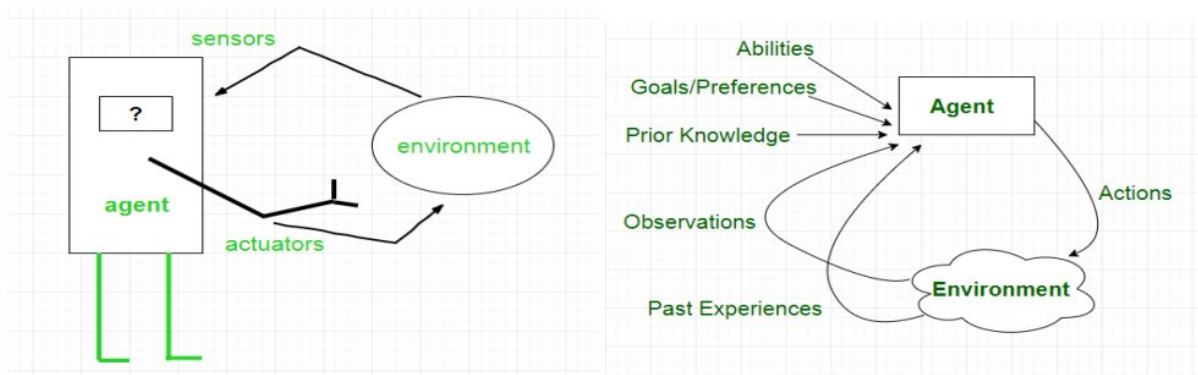
Deep learning, big data and artificial general intelligence (2011-present)

1. **Year 2011:** In the year 2011, IBM's Watson won jeopardy, a quiz show, where it had to solve the complex questions as well as riddles. Watson had proved that it could understand natural language and can solve tricky questions quickly.
2. **Year 2012:** Google has launched an Android app feature "Google now", which was able to provide information to the user as a prediction.
3. **Year 2014:** In the year 2014, Chatbot "Eugene Goostman" won a competition in the infamous "Turing test."
4. **Year 2018:** The "Project Debater" from IBM debated on complex topics with two master debaters and also performed extremely well.
5. Google has demonstrated an AI program "Duplex" which was a virtual assistant and which had taken hairdresser appointment on call, and lady on other side didn't notice that she was talking with the machine.

### Agents and environments:

An agent is anything that can be viewed as perceiving its environment through sensors and sensor acting upon that environment through actuators.

1. A human agent has eyes, ears, and other organs for sensors and hands, legs, mouth, and other body parts for actuators.
2. A robotic agent might have cameras and infrared range finders for sensors and various motors for actuators.
3. A software agent receives keystrokes, file contents, and network packets as sensory inputs and acts on the environment by displaying on the screen, writing files, and sending network packets.



**FIG:AGENT WORKING**

**Sensor:** Sensor is a device which detects the change in the environment and sends the information to other electronic devices. An agent observes its environment through sensors.

**Actuators:** Actuators are the component of machines that converts energy into motion. The actuators are only responsible for moving and controlling a system. An actuator can be an electric motor, gears, rails, etc.

**Effectors:** Effectors are the devices which affect the environment. Effectors can be legs, wheels, arms, fingers, wings, fins, and display screen.

### **Percept**

We use the term **percept** to refer to the agent's perceptual inputs at any given instant.

#### **Percept Sequence**

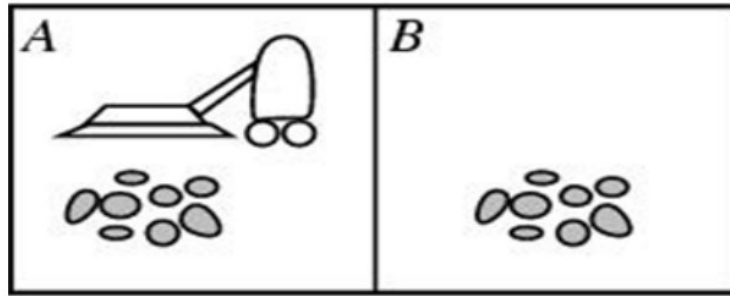
An agent's **percept sequence** is the complete history of everything the agent has ever perceived.

### **Agent function**

Mathematically speaking, we say that an agent's behavior is described by the **agent function** that maps any given percept sequence to an action.

### **Agent program**

1. The agent function for an artificial agent will be implemented by an agent program.
2. It is important to keep these two ideas distinct.
3. The agent function is an abstract mathematical description;
4. the agent program is a concrete implementation, running on the agent architecture.
5. To illustrate these ideas, we will use a very simple example-the vacuum-cleaner world shown in Figure.
6. This particular world has just two locations: squares A and B.
7. The vacuum agent perceives which square it is in and whether there is dirt in square.
8. It can choose to move left, move right, suck up the dirt, or do nothing.
9. One very simple agent function is the following:
10. if the current square is dirty, then suck, otherwise,
11. it move to the other square.
12. A partial tabulation of this agent function is shown in Figure.



Agent function

| Percept Sequence       | Action |
|------------------------|--------|
| [A, Clean]             | Right  |
| [A, Dirty]             | Suck   |
| [B, Clean]             | Left   |
| [B, Dirty]             | Suck   |
| [A, Clean], [A, Clean] | Right  |
| [A, Clean], [A, Dirty] | Suck   |
| .....                  | .....  |

Agent program

```

function Reflex-VACUUM-AGENT ([locations, status]) returns an
    action if status = Dirty then return Suck
    else if location = A then return Right
    elseif location = B then return Left
    
```

**Intelligent Agents:**

An intelligent agent is an autonomous entity which act upon an environment using sensors and actuators for achieving goals. An intelligent agent may learn from the environment to achieve their goals. A thermostat is an example of an intelligent agent.

Following are the main four rules for an AI agent:

- **Rule 1:** An AI agent must have the ability to perceive the environment.
- **Rule 2:** The observation must be used to make decisions.
- **Rule 3:** Decision should result in an action.
- **Rule 4:** The action taken by an AI agent must be a rational action.

Structure of an AI Agent

The task of AI is to design an agent program which implements the agent function. The structure of an intelligent agent is a combination of architecture and agent program. It can be viewed as:

$$\text{Agent} = \text{Architecture} + \text{Agent program}$$

Following are the main three terms involved in the structure of an AI agent:

**Architecture:** Architecture is machinery that an AI agent executes on.

**Agent Function:** Agent function is used to map a percept to an action.

**Agent program:** Agent program is an implementation of agent function. An agent program executes on the physical architecture to produce function  $f$ .

### **Rational Agent:**

A rational agent is an agent which has clear preference, models uncertainty, and acts in a way to maximize its performance measure with all possible actions.

A rational agent is said to perform the right things. AI is about creating rational agents to use for game theory and decision theory for various real-world scenarios.

For an AI agent, the rational action is most important because in AI reinforcement learning algorithm, for each best possible action, agent gets the positive reward and for each wrong action, an agent gets a negative reward.

### **Good Behavior: The concept of Rationality**

Rationality:

The rationality of an agent is measured by its performance measure. Rationality can be judged on the basis of following points:

- Performance measure which defines the success criterion.
- Agent prior knowledge of its environment.
- Best possible actions that an agent can perform.
- The sequence of percepts.

### **Performance measures**

1. A performance measure embodies the criterion for success of an agent's behavior.
2. When an agent is plunked down in an environment, it generates a sequence of ~~acts~~ actions according to the percepts it receives.
3. This sequence of actions causes the environment to go through a sequence of states.
4. If the sequence is desirable, then the agent has performed well.

## PEAS Representation

PEAS is a type of model on which an AI agent works upon. When we define an AI agent or rational agent, then we can group its properties under PEAS representation model. It is made up of four words:

- **P:** Performance measure
- **E:** Environment
- **A:** Actuators
- **S:** Sensors

### Examples:

#### 1) Self Driving car

**Performance:** Safety, time, legal drive, comfort

**Environment:** Roads, other vehicles, road signs, pedestrian

**Actuators:** Steering, accelerator, brake, signal, horn

**Sensors:** Camera, GPS, speedometer, odometer, accelerometer, sonar.

#### 2) Medical Diagnose

**Performance:** Healthy patient, Minimized cost

**Environment:** Patient, Hospital, Staff

**Actuators:** Tests, Treatments

**Sensors:** Keyboard (Entry of symptoms)

#### 3) Vacuum Cleaner

**Performance:** Cleanness, Efficiency, Battery life, Security

**Environment:** Patient, Hospital, Staff

**Actuators:** Wheels, Brushes, Vacuum, Extractor

**Sensors:** Camera, Dirt detection sensor, Cliff sensor, Bump Sensor, Infrared Wall Sensor

#### 4) Part -picking Robot

**Performance:** Percentage of parts in correct bins.

**Environment:** Conveyor belt with parts, Bins

**Actuators:** Jointed Arms, Hand

**Sensors:** Camera, Joint angle sensors.

## Types of Agents

Agents can be grouped into four classes based on their degree of perceived intelligence and capability :

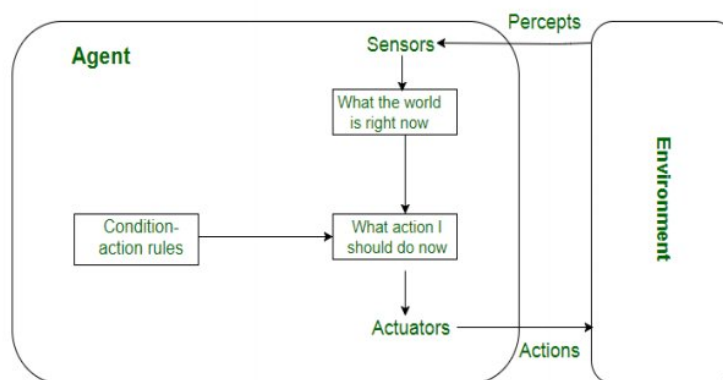
- Simple Reflex Agents
- Model-Based Reflex Agents
- Goal-Based Agents
- Utility-Based Agents
- Learning Agent

### Simple reflex agents

Simple reflex agents ignore the rest of the percept history and act only on the basis of the **current percept**. Percept history is the history of all that an agent has perceived to date. The agent function is based on the **condition-action rule**. A condition-action rule is a rule that maps a state i.e, condition to an action. If the condition is true, then the action is taken, else not. This agent function only succeeds when the environment is fully observable. For simple reflex agents operating in partially observable environments, infinite loops are often unavoidable. It may be possible to escape from infinite loops if the agent can randomize its actions.

Problems with Simple reflex agents are :

- Very limited intelligence.
- No knowledge of non-perceptual parts of the state.
- Usually too big to generate and store.
- If there occurs any change in the environment, then the collection of rules need to be updated.



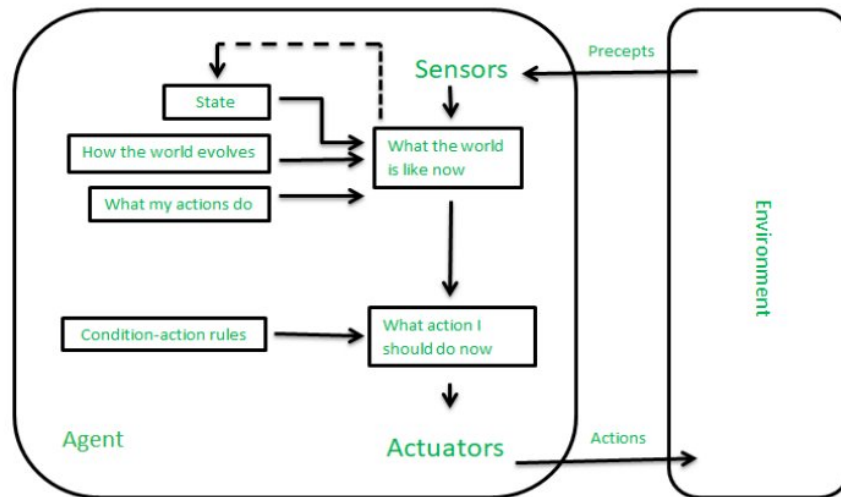
### Model-based reflex agents

It works by finding a rule whose condition matches the current situation. A model-based agent can handle **partially observable environments** by the use of a model about the world. The agent has to keep track

of the **internal state** which is adjusted by each percept and that depends on the percept history. The current state is stored inside the agent which maintains some kind of structure describing the part of the world which cannot be seen.

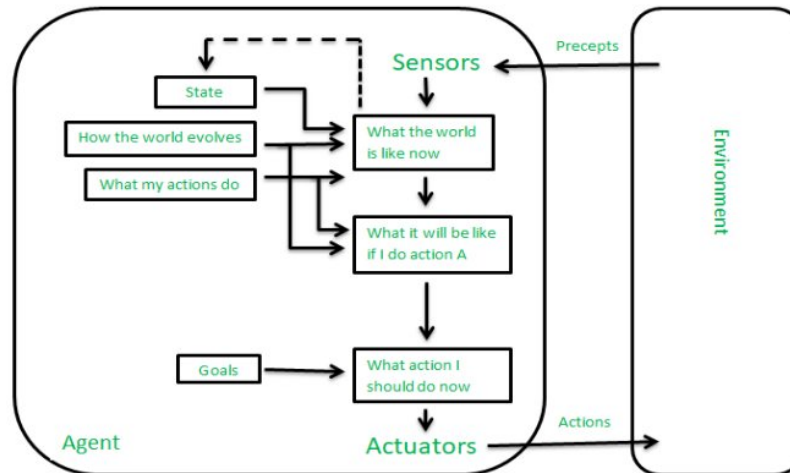
Updating the state requires information about :

- how the world evolves independently from the agent, and
- how the agent's actions affect the world.



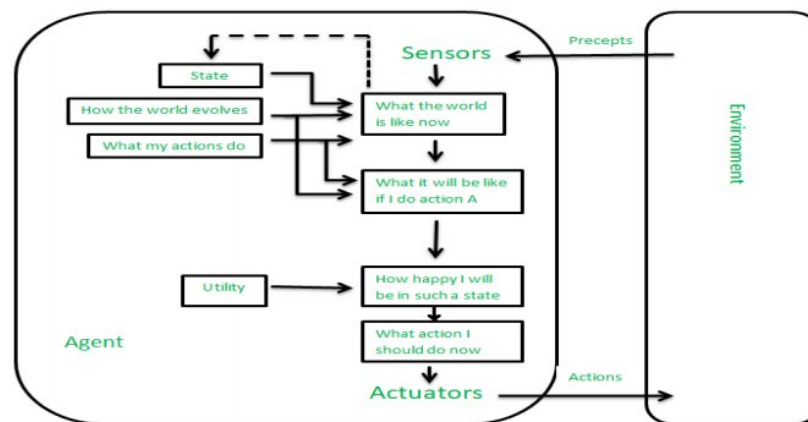
## Goal-based agents

These kinds of agents take decisions based on how far they are currently from their **goal**(description of desirable situations). Their every action is intended to reduce its distance from the goal. This allows the agent a way to choose among multiple possibilities, selecting the one which reaches a goal state. The knowledge that supports its decisions is represented explicitly and can be modified, which makes these agents more flexible. They usually require search and planning. The goal-based agent's behavior can easily be changed.



## Utility-based agents

The agents which are developed having their end uses as building blocks are called utility-based agents. When there are multiple possible alternatives, then to decide which one is best, utility-based agents are used. They choose actions based on a **preference (utility)** for each state. Sometimes achieving the desired goal is not enough. We may look for a quicker, safer, cheaper trip to reach a destination. Agent happiness should be taken into consideration. Utility describes how **“happy”** the agent is. Because of the uncertainty in the world, a utility agent chooses the action that maximizes the expected utility. A utility function maps a state onto a real number which describes the associated degree of happiness.



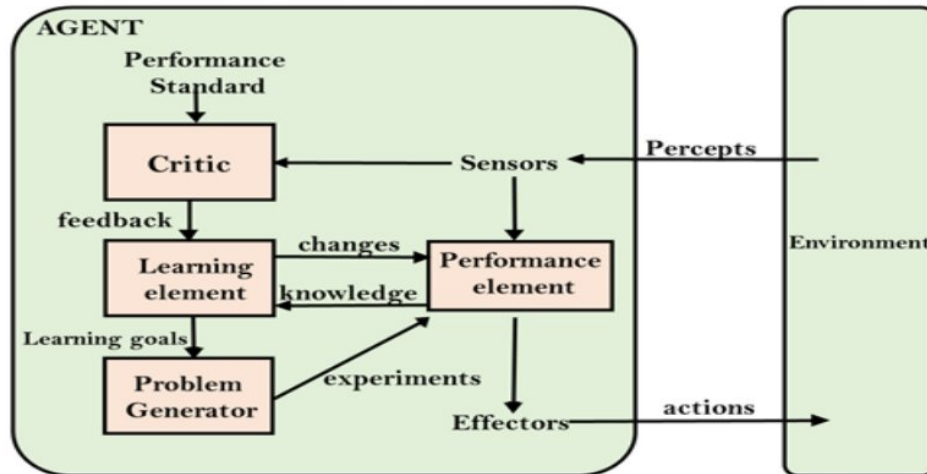
## Learning Agent :

A learning agent in AI is the type of agent that can learn from its past experiences or it has learning capabilities. It starts to act with basic knowledge and then is able to act and adapt automatically through learning.

A learning agent has mainly four conceptual components, which are:

1. **Learning element:** It is responsible for making improvements by learning from the environment

2. **Critic:** The learning element takes feedback from critics which describes how well the agent is doing with respect to a fixed performance standard.
3. **Performance element:** It is responsible for selecting external action
4. **Problem Generator:** This component is responsible for suggesting actions that will lead to new and informative experiences.



## The Nature of Environments

An environment is everything in the world which surrounds the agent, but it is not a part of an agent itself. An environment can be described as a situation in which an agent is present.

The environment is where agent lives, operate and provide the agent with something to sense and act upon it. An environment is mostly said to be non-feministic.

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## Features of Environment

An environment can have various features from the point of view of an agent:

1. Fully observable vs Partially Observable
2. Static vs Dynamic
3. Discrete vs Continuous
4. Deterministic vs Stochastic
5. Single-agent vs Multi-agent
6. Episodic vs sequential
7. Known vs Unknown
8. Accessible vs Inaccessible

1. Fully observable vs Partially Observable:

1. If an agent sensor can sense or access the complete state of an environment at each point of time then it is a **fully observable** environment, else it is **partially observable**.
2. A fully observable environment is easy as there is no need to maintain the internal state to keep track history of the world.
3. An agent with no sensors in all environments then such an environment is called as **unobservable**.

2. Deterministic vs Stochastic:

1. If an agent's current state and selected action can completely determine the next state of the environment, then such environment is called a deterministic environment.
2. A stochastic environment is random in nature and cannot be determined completely by an agent.
3. In a deterministic, fully observable environment, agent does not need to worry about uncertainty.

3. Episodic vs Sequential:

1. In an episodic environment, there is a series of one-shot actions, and only the current percept is required for the action.
2. However, in Sequential environment, an agent requires memory of past actions to determine the next best actions

#### 4. Single-agent vs Multi-agent

1. If only one agent is involved in an environment, and operating by itself then such an environment is called single agent environment.
2. However, if multiple agents are operating in an environment, then such an environment is called a multi-agent environment.
3. The agent design problems in the multi-agent environment are different from single agent environment.

#### 5. Static vs Dynamic:

1. If the environment can change itself while an agent is deliberating then such environment is called a dynamic environment else it is called a static environment.
2. Static environments are easy to deal because an agent does not need to continue looking at the world while deciding for an action.
3. However for dynamic environment, agents need to keep looking at the world at each action.
4. Taxi driving is an example of a dynamic environment whereas Crossword puzzles are an example of a static environment.

#### 6. Discrete vs Continuous:

1. If in an environment there are a finite number of percepts and actions that can be performed within it, then such an environment is called a discrete environment else it is called continuous environment.
2. A chess game comes under discrete environment as there is a finite number of moves that can be performed.
3. A self-driving car is an example of a continuous environment.

#### 7. Known vs Unknown

1. Known and unknown are not actually a feature of an environment, but it is an agent's state of knowledge to perform an action.
2. In a known environment, the results for all actions are known to the agent. While in unknown environment, agent needs to learn how it works in order to perform an action.
3. It is quite possible that a known environment to be partially observable and an Unknown environment to be fully observable.

#### 8. Accessible vs Inaccessible

1. If an agent can obtain complete and accurate information about the state's environment, then such an environment is called an Accessible environment else it is called inaccessible.
2. An empty room whose state can be defined by its temperature is an example of an accessible environment.
3. Information about an event on earth is an example of Inaccessible environment.

## 1.4 THE STATE OF THE ART

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What can AI do today? A concise answer is difficult because there are so many activities in so many subfields. Here we sample a few applications; others appear throughout the book.

**Robotic vehicles:** A driverless robotic car named STANLEY sped through the rough terrain of the Mojave dessert at 22 mph, finishing the 132-mile course first to win the 2005 DARPA Grand Challenge. STANLEY is a Volkswagen Touareg outfitted with cameras, radar, and laser rangefinders to sense the environment and onboard software to command the steering, braking, and acceleration (Thrun, 2006). The following year CMU's BOSS won the Urban Challenge, safely driving in traffic through the streets of a closed Air Force base, obeying traffic rules and avoiding pedestrians and other vehicles.

**Speech recognition:** A traveler calling United Airlines to book a flight can have the entire conversation guided by an automated speech recognition and dialog management system.

**Autonomous planning and scheduling:** A hundred million miles from Earth, NASA's Remote Agent program became the first on-board autonomous planning program to control the scheduling of operations for a spacecraft (Jonsson *et al.*, 2000). REMOTE AGENT generated plans from high-level goals specified from the ground and monitored the execution of those plans—detecting, diagnosing, and recovering from problems as they occurred. Successor program MAPGEN (Al-Chang *et al.*, 2004) plans the daily operations for NASA's Mars Exploration Rovers, and MEXAR2 (Cesta *et al.*, 2007) did mission planning—both logistics and science planning—for the European Space Agency's Mars Express mission in 2008.

**Game playing:** IBM's DEEP BLUE became the first computer program to defeat the world champion in a chess match when it bested Garry Kasparov by a score of 3.5 to 2.5 in an exhibition match (Goodman and Keene, 1997). Kasparov said that he felt a "new kind of intelligence" across the board from him. *Newsweek* magazine described the match as "The brain's last stand." The value of IBM's stock increased by \$18 billion. Human champions studied Kasparov's loss and were able to draw a few matches in subsequent years, but the most recent human-computer matches have been won convincingly by the computer.

**Spam fighting:** Each day, learning algorithms classify over a billion messages as spam, saving the recipient from having to waste time deleting what, for many users, could comprise 80% or 90% of all messages, if not classified away by algorithms. Because the spammers are continually updating their tactics, it is difficult for a static programmed approach to keep up, and learning algorithms work best (Sahami *et al.*, 1998; Goodman and Heckerman, 2004).

**Logistics planning:** During the Persian Gulf crisis of 1991, U.S. forces deployed a Dynamic Analysis and Replanning Tool, DART (Cross and Walker, 1994), to do automated logistics planning and scheduling for transportation. This involved up to 50,000 vehicles, cargo, and people at a time, and had to account for starting points, destinations, routes, and conflict resolution among all parameters. The AI planning techniques generated in hours a plan that would have taken weeks with older methods. The Defense Advanced Research Project Agency (DARPA) stated that this single application more than paid back DARPA's 30-year investment in AI.

**Robotics:** The iRobot Corporation has sold over two million Roomba robotic vacuum cleaners for home use. The company also deploys the more rugged PackBot to Iraq and Afghanistan, where it is used to handle hazardous materials, clear explosives, and identify the location of snipers.

**Machine Translation:** A computer program automatically translates from Arabic to English, allowing an English speaker to see the headline "Ardogan Confirms That Turkey Would Not Accept Any Pressure, Urging Them to Recognize Cyprus." The program uses a statistical model built from examples of Arabic-to-English translations and from examples of English text totaling two trillion words (Brants *et al.*, 2007). None of the computer scientists on the team speak Arabic, but they do understand statistics and machine learning algorithms.

These are just a few examples of artificial intelligence systems that exist today. Not magic or science fiction—but rather science, engineering, and mathematics, to which this book provides an introduction.